

ISTA 410/INFO 510 Bayesian Modelling and Inference

Homework 3

Total points: 20

Submission: Submit your answers in a single PDF file.

Q.1. (9 points; 3 points for each part)

In 1998, the New York Times and CBS News polled 1048 randomly selected 13 – 17 year olds to ask them if they had a television in their room. Among this group of teenagers, 692 of them said they had a television in their room. Alex and Benedict both want to use the binomial model for this dataset, but they have different prior beliefs about p , the proportion of teenagers having a television in their room

- a) Alex asks 10 friends the same question, and 8 of them have a television in their room. Alex decides to use this information to construct his prior. Design a continuous beta prior reflecting Alex's belief.
- b) Calculate Alex's posterior and Benedict's posterior distributions. Plot the two priors on one graph, and plot the corresponding posteriors on another graph. [Hint – Benedict's prior, $p \sim \text{Beta}(25.74, 43.06)$]
*Optional - Obtain 95% credible intervals for Alex and Benedict.
- c) Conduct prior predictive checks for Alex and Benedict. Which prior (Alex's or Benedict's) do you think is more appropriate for this teenagers and television data. Explain.

Q.2. (11 points; 5 points for sub-question a, and 2 points each for the "Additional questions" in b)

- a) There is a "Markov-exercise.txt" file that is uploaded on D2L, run the R script (it's a R markdown script) and explain the key insights and conclusion from this analysis.
- b) Solve the three additional questions at the end of the script and show the output for each of them.